

RESPIRATORY SYSTEM

1. The physiological dead space:

- a) Is exactly equal to the anatomic dead space in normal people
- b) Increases during exercise
- c) Is measured by single breath nitrogen analysis
- d)** Is higher in elderly than young adults
- e) Is the same as the physiological shunt

2. A high alveolo-arterial PO₂ difference most likely results from:

- a) Low fractional concentration of O₂ in the inspired air
- b) Hypoventilation
- c) Arteriovenous shunt
- d)** Pulmonary edema
- e) Polycythemia

3. At 33 feet under water the ambient pressure is:

- a) Two atmospheres
- b) One and a half atmosphere
- c) 33 pounds per square inch
- d)** Same as the pressure in a tunnel 33 feet below sea level
- e) Half the pressure 33 feet above sea level

4. Which of the following combinations are typical findings in compensated respiratory alkalosis:

- a) Low PaCO₂, low bicarbonate level and normal pH
- b) Low PaCO₂, normal bicarbonate level and high pH
- c) Low PaCO₂, low bicarbonate level and high pH
- d)** Low PaCO₂, normal bicarbonate level and normal pH
- e) Low PaCO₂, high bicarbonate level and high pH

5. The partial pressure of oxygen in the alveoli is expected to be least affected by:

- a) Fractional concentration of O₂ in inspired air
- b) Respiratory quotient
- c)** PCO₂ in alveoli
- d) Hemoglobin concentration
- e) Barometric pressure

6. A veno-arterial shunt is likely to:

- a) Decrease the cardiac output
- b) Increase the pulse pressure
- c) Increase venous return
- d)** Decrease PaO₂
- e) Increase the peripheral vascular resistance

7. The arterio-venous difference of O₂ is highest across:

- a)** Liver
- b) Kidney
- c) Cardiac muscle
- d) Skin
- e) Brain

8. In a patient, hypoventilation is most likely if:

- a) PaO₂ is low
- b) pH of arterial blood is low
- c)** PaCO₂ is high
- d) O₂ in arterial blood is low

e) PCO₂ of expired air is high

9. A decrease in the recoil force of the lung is likely to:

- a) Decrease total lung capacity
- b) Increase vital capacity
- c) Increase functional residual capacity
- d) Decrease the residual volume
- e) Decrease compliance of the lung

10. Central chemoreceptors differ from chemoreceptors in that they:

- a) Respond to changes in PaCO₂
- b) Respond to changes in pH
- c) Do not respond to changes in PO₂
- d) Become more sensitive to CO₂ with chronic exposure
- e) They are less sensitive to changes in PaCO₂ than pH

11. Surfactant:

- a) Helps to equalize pressure within interconnected alveoli
- b) Reduces surface tension more effectively as alveolus becomes smaller
- c) Increases lung compliance
- d) Increases the slope of the pressure-volume curve
- e) All of the above

12. One month of an exposure to an inspired CO₂ of 5% would result in:

- a) Increased ventilatory sensitivity to CO₂
- b) Reduced ventilatory sensitivity to hypoxia
- c) Increased plasma bicarbonate
- d) Alkalosis in arterial plasma
- e) Increased alveolar PO₂

13. In a normal adult man, breathing 100% oxygen:

- a) Is likely to double O₂ in arterial blood
- b) Is likely to reduce alveolar PCO₂
- c) Is likely to raise PO₂ in venous blood to about 200 mmHg
- d) Is likely to raise arterial PO₂ to about 660 mmHg
- e) Increases the affinity of hemoglobin to oxygen

14. At the end of maximum inspiration:

- a) Intra-alveolar pressure is zero
- b) Intra-pleural pressure is zero
- c) Recoil force of the chest is greater than the recoil force of the lung
- d) Surface tension is lowest
- e) Average alveolar PO₂ is lower than at the end of maximum expiration

15. On ascent to a high altitude:

- a) The concentration of oxygen in the air decreases
- b) The number of oxygen molecules per liter of air decreases
- c) Alveolar PCO₂ is kept constant despite the high ventilation
- d) Density of air increases
- e) The PCO₂ in air is the same as at sea level

16. Compliance of the lung:

- a) Is higher in small children than in adults
- b) Is about 500 ml/cm H₂O
- c) Is normally twice the compliance of the chest
- d) Is higher in elderly than young adults
- e) Is decreased in emphysematous changes

17. Denervation of carotid and aortic bodies leads to all the following except:

- a) Decreased sensitivity of ventilation to change in PCO₂
- b) Decreased sensitivity of ventilation to change in pH
- c) Complete absence of response of ventilation to change in PO₂
- d) Absence response of ventilation to exercise
- e) Decreased sensitivity of ventilaton to stagnant hypoxia

18. Exchange of which of the following gases across the respiratory membrane is normally diffusion limited:

- a) Oxygen
- b) Carbon dioxide
- c) Carbon monoxide
- d) Nitrous oxide
- e) Nitrogen

19. Minute ventilation is:

- a) The volume of air moved in or out of the alveoli per minute
- b) The volume of air moved in or out of the lung per breath
- c) Alveolar ventilation plus dead space ventilation
- d) Invariably increases if the respiratory rate increases
- e) Is measured using Bohr's equation (CO₂ analysis in expired air)

20. The ventilatory response to hypoxia:

- a) Is independent of arterial PCO₂
- b) Is mediated by the central chemoreceptors
- c) Decreases gradually with time
- d) Is largely mediated by the aortic bodies in humans
- e) None of the above

21. With respect to oxygen and carbon dioxide transport in the blood:

- a) For the same partial pressure a unit of blood carries the same amount of O₂ and CO₂
- b) High PCO₂ favors O₂ binding to hemoglobin
- c) Metabolic acidosis reduces PCO₂
- d) Oxygenation of hemoglobin increases the affinity of hemoglobin to CO₂
- e) Per unit volume of arterial blood there is more O₂ than CO₂

22. PCO₂ in arterial blood:

- a) Invariably increases with hypoxia
- b) Stimulates ventilation mainly through peripheral chemoreceptors
- c) Is the major controller of ventilation
- d) Increases early in exercise
- e) Increases on ascent to high altitude

23. The functional residual capacity:

- a) Is the volume of air in the lung at the end of normal inspiration
- b) Increases with age
- c) Is measured by simple spirometry
- d) Is about 3 liters in an average adult male
- e) Is about 1 liter in average adult male

24. At the end of inspiration at sea level the PO₂ in the anatomic dead space is approximately:

- a) 150 mmHg
- b) 100 mmHg
- c) 160 mmHg

- d) 95 mmHg
- e) Above a hundred but less than 150 mmHg

25. Which is true concerning control of ventilation:

- a) Ventilation is more sensitive to changes in PO_2 than changes in PCO_2 in arterial blood
- b) Carbon dioxide stimulates ventilation only through the central chemoreceptors
- c) At high altitude the primary stimulus to ventilation is high PCO_2
- d) The response of ventilation to a sustained rise in PCO_2 increases with time
- e) Cutting the vagus nerves decreases the depth of breathing

26. Which of the following is likely to decrease airway resistance?

- a) Beta-adrenergic blockers
- b) Muscarinic cholinergic agonists
- c) Breathing at higher lung volumes
- d) Leukotrienes
- e) Histamine

27. Compared to normal arterial blood, normal mixed venous blood has:

- a) Higher PO_2 , lower PCO_2 and higher pH
- b) Lower PO_2 , higher PCO_2 and higher pH
- c) Lower PO_2 , higher PCO_2 and lower pH
- d) Lower PO_2 , lower PCO_2 and higher pH
- e) Higher PO_2 , lower PCO_2 and lower pH

28. Deficiency of surfactant is likely to result in:

- a) Increased compliance
- b) Decreased work of breathing
- c) Reduced surface tension of fluid in the alveoli
- d) Decreased compliance
- e) Asthma

29. For a normal Hb- O_2 dissociation curve, the most correct relationship is:

- a) PO_2 is 40 mmHg, percent saturation 50
- b) PO_2 is 95 mmHg, percent saturation 97
- c) PO_2 is 30 mmHg, percent saturation 50
- d) PO_2 is 60 mmHg, percent saturation 60
- e) PO_2 is 50 mmHg, percent saturation 50

30. In a normal subject, blood at the end of the pulmonary capillary:

- a) Has a PO_2 similar to blood in the left ventricle
- b) Has a PO_2 similar to that in the left atrium
- c) Has a PO_2 higher than blood from the aorta
- d) Has a percent saturation of 95
- e) Contains 20 mL of CO_2 per 100 mL

31. PCO_2 in arterial blood:

- a) Invariably increases with hypoxia
- b) Stimulates ventilation mainly through peripheral chemoreceptors
- c) Is lower than normal in compensated metabolic acidosis
- d) Increases early in exercise
- e) Increases on ascent to high altitude

32. In which of the following conditions is the percent saturation of hemoglobin in arterial blood likely to be normal :

- a) Venoarterial shunts
- b) Pulmonary edema
- c) Carbon monoxide poisoning
- d) Anemia
- e) Methemoglobinemia

33. At which of the following points is the intrapleural pressure closest to zero:

- a) End of normal expiration
- b) End of normal inspiration
- c) End of maximum expiration
- d) End of maximum inspiration
- e) Mid-normal expiration

34. Regarding the vital capacity, all of the following are correct except:

- a) It is higher in young adults than small children
- b) It is higher in males than females
- c) It is normal in pure restrictive lung disease
- d) It can be measured by simple spirometry
- e) It is related to body size

35. Immediate acclimatization to high altitude is by:

- a) Polycythemia
- b) Hyperventilation
- c) Increased cardiac output
- d) Shift of hemoglobin dissociation curve to the left
- e) Increase formation of fetal hemoglobin

36. At the end of maximal expiration the volume of air in the lungs is:

- a) FRC - RV
- b) IRV + RV
- c) FRC - TV
- d) TLC - VC
- e) VC - ERV - IRV

37. At the end of expiration at sea level the PO₂ in the anatomic dead space is approximately:

- a) 150 mmHg
- b) 100 mmHg
- c) 160 mmHg
- d) 40 mmHg
- e) 130 mmHg

38. Which is true regarding lung and chest wall mechanics:

- a) At maximum expiration the lungs have no tendency to collapse but the chest has a great tendency to expand
- b) At end of normal inspiration the recoil tendency of the lung is greater than that of the chest
- c) During inspiration the recoil tendencies of both the lungs and chest increases
- d) At the end of normal expiration the recoil tendency of the chest is slightly greater than that of the lung
- e) The combined compliance is greater than the individual compliances of the lung and chest

39. Which is true concerning the diffusing capacity of the lungs:

- a) It increases whenever the partial pressure gradient increases
- b) It increases as the alveolar capillary surface area rises

- c) It increases as the thickness of the respiratory membrane area rises
- d) It is usually measured using carbon dioxide
- e) It is usually measured using oxygen

40. A person whose anatomic dead space is 100 mL breathes 12 times per minute with a tidal volume of 400 mL. His pulmonary ventilation is:

- a) 1.2 liters
- b) 2.4 liters
- c) 3.6 liters
- d) 4.8 liters**
- e) 6.0 liters

41. While repaying the O₂ dept:

- a) The [H⁺] is increasing
- b) PCO₂ is decreasing
- c) Muscle blood flow is increasing
- d) Respiratory rate is increasing**
- e) Minute volume is decreasing

42. An average adult male has a vital capacity of:

- a) 3 liters
- b) 4 liters
- c) 5 liters**
- d) 6 liters
- e) 7 liters

43. The residual volume can be calculated by subtracting the expiratory reserve volume from:

- a) Vital Capacity
- b) Inspiratory capacity
- c) Functional residual capacity
- d) Total lung capacity
- e) Alveolar ventilation

44. The volume of air that describes the lung's ability to eliminate carbon dioxide is:

- a) Tidal volume
- b) Pulmonary ventilation
- c) Vital capacity**
- d) Alveolar ventilation
- e) Functional residual capacity

45. A decrease in the recoil force of the lung is expected to:

- a) Increase vital capacity
- b) Decrease residual volume
- c) Decrease compliance of the lung
- d) Increase functional residual capacity
- e) Decrease total lung capacity

46. Deficiency of surfactant may lead to:

- a) Increased work of breathing**
- b) Increased compliance of the lung
- c) Decreased recoil force of the lung
- d) Bronchial asthma
- e) An increase in functional residual capacity

47. Continuous measurement of the concentration of nitrogen in the expired air can be used to measure:

- a) Physiological dead space
- b) Alveolar ventilation
- c) Anatomic dead space
- d) Functional residual capacity
- e) Expiratory reserve volume

48. Oxygen consumption of an average adult man per minute is about:

- a) 250 ml
- b) 350 ml
- c) 400 ml
- d) 500 ml
- e) 550 ml

49. Which of the following is expected to increase function residual capacity:

- a) Surfactant deficiency
- b) Lung fibrosis
- c) Degeneration of lung elastic tissue
- d) Pulmonary edema
- e) Pulmonary congestion

50. The most likely cause of hypoxic hypoxia together with hypercapnia is:

- a) Ascent to a high altitude
- b) Mild pulmonary edema
- c) Veno-arterial shunt
- d) Hypoventilation
- e) Slight ventilation perfusion mismatch

51. Which of the following is correct regarding blood in the pulmonary trunk:

- a) PO_2 of 60 mmHg
- b) 50% saturation of hemoglobin
- c) PCO_2 of 40 mmHg
- d) Bicarbonate concentration lower than in the aorta
- e) PN_2 higher than in aorta

52. Ventilation is most sensitive to changes in:

- a) PO_2
- b) pH of arterial blood
- c) PCO_2
- d) Body temperature
- e) % saturation of hemoglobin

53. Inspiration is initiated by discharge from:

- a) The dorsal inspiratory group of neurons
- b) The ventral group of neurons
- c) The preBottzinger complex
- d) Apneustic center
- e) Pneumotaxic center

54. Voluntary hyperventilation by a normal person is expected to lead to all the following *except*:

- a) Alkalosis
- b) Hypocapnia
- c) Low blood bicarbonate concentration in arterial blood
- d) A great increase in oxygen concentration in arterial blood

e) Low ionized plasma calcium

55. The normal diffusing capacity of the respiratory membrane of an average adult male for oxygen is about:

- a) 10 ml
- b) 25 ml
- c) 50 ml
- d) 4 ml
- e) 15 ml

56. Cyanosis is unlikely to occur in:

- a) Pulmonary edema
- b) Hypoventilation
- c) Severe anemia
- d) Radiation
- e) Convection

57. Which is true concerning the diffusing capacity of the lungs:

- a) Rises as the partial pressure gradient rises
- b) Rises as the effective surface area of the respiratory membrane rises
- c) Is not affected by the thickness of the respiratory membrane
- d) Is measured using radioactive CO₂
- e) Is not affected by the diffusion coefficient of the membrane to gases

58. Which type of hypoxia in which there is a greater than normal atrial-mixed venous partial pressure difference and relatively normal arterial oxygen concentration:

- a) Hypoxic hypoxia
- b) Anemic hypoxia
- c) Stagnant hypoxia
- d) Histotoxic hypoxia
- e) B and C are correct

59. Administration of a gas mixture with higher than normal oxygen concentration at atmospheric pressure would elevate hypoxia due to:

- a) Veno-arterial shunt
- b) Diffusion impairment
- c) Anemia
- d) Cyanide poisoning
- e) Reduced blood flow to tissues

60. The PCO₂ in an alveolus which is ventilated but not perfused is about:

- a) 0.3 mmHg
- b) 46 mmHg
- c) 40 mmHg
- d) 5 mmHg
- e) 7 mmHg

61. Which is true concerning surface tension in lungs?

- a) The normal amount of surfactant reduces surface tension in healthy lungs to a negligible level
- b) In the absence of surfactant alveoli with small radii collapse into alveoli with big radii
- c) Surfactant is maximally concentrated at liquid-air interface at the end of inspiration
- d) In normal lungs surface tension is responsible for 20% of the recoil force of the lung

e) Surface tension remains the same throughout the breathing cycle

62. Which is true concerning oxygen transport in the blood?

- a) If PO_2 is 60 mmHg oxygen concentration is about 18 ml/100 ml blood
- b) The P_{50} is the percent saturation of hemoglobin with oxygen when PO_2 is 50 mmHg
- c) The normal P_{50} of arterial blood is higher than that of venous blood
- d) Increase in pH increases the P_{50}
- e) P_{50} reflects the capacity of blood to carry oxygen

63. The physiological shunt:

- a) Represents the wasted perfusion
- b) Is equal to the physiological dead space
- c) Is measured by single breath nitrogen analysis
- d) Occurs only in people with respiratory diseases
- e) Is about 5% of the cardiac output

64. At the summit of a mountain the barometric pressure was found to be 300 mmHg. The partial pressure of O_2 in the air is saturated with water vapour at $37^\circ C$ at this altitude is expected to be about:

- a) 45 mmHg
- b) 55 mmHg
- c) 60 mmHg
- d) 65 mmHg
- e) 62 mmHg

65. Which of the following parameters is lower in the elderly than in young adults?

- a) Residual volume
- b) Vital capacity
- c) Lung compliance
- d) Functional residual capacity
- e) Physiological dead space

66. Surfactant:

- a) Is produced by type I pneumocytes
- b) Deficiency decreases lung compliance
- c) Deficiency decreases work of breathing
- d) Production starts only after birth in full term babies
- e) Production is inhibited by glucocorticoids

67. At the end of normal expiration all the following are correct *except*:

- a) Recoil tendency of the lung as acting inward to collapse the lung
- b) Chest recoil is acting outward to expand the lung
- c) Alveolar pressure is greater than mouth pressure
- d) The average intrapleural pressure is zero
- e) Surface tension is less than at the end of normal inspiration

68. Obstructive lung disease is typically characterized by:

- a) Reduced total lung capacity
- b) Reduced FEV1
- c) Reduced functional residual capacity
- d) Slightly increased FEV1/FVC ratio
- e) Reduced residual volume

69. The following parameters are higher in the elderly than young adults:

- a) Vital capacity
- b) Basal oxygen consumption per kg body weight
- c) Functional residual capacity
- d) Anatomical dead space
- e) Peak expiratory flow

70. The approximate dead space of a normal adult man breathing through a tube that is 50 cm long and 6mm in diameter is:

- a) 200 ml
- b) 300 ml
- c) 150 ml
- d) 250 ml
- e) 350 ml

71. Spontaneous respiration ceases after:

- a) Transection of the spinal cord below C5
- b) Transection of the brain stem between the pons and medulla combined with bilateral vagotomy
- c) Transection of the spinal cord
- d) Complete destruction of the peripheral chemoreceptors
- e) Transection of the spinal cord at the first cervical segment

72. All the following are expected to decrease airway resistance *except*:

- a) Atropine
- b) Adrenaline
- c) Parasympathetic stimulation
- d) Sympathetic stimulation
- e) Shifting from nose to mouth breathing